

# Uranium and carbon isotopes document global-ocean redox-productivity relationships linked to cooling during the Frasnian-Famennian mass extinction

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## ABSTRACT

The cause of the Frasnian-Famennian boundary (FFB) biotic crisis, one of the “Big Five” Phanerozoic mass extinctions, remains poorly understood. Here, we generated a high-resolution uranium-isotope profile ( $\delta^{238}\text{U}$ ) for a marine carbonate section at Baisha, South China, in order to document secular variation in mean global-ocean redox conditions and to compare its relationship to coeval changes in organic carbon burial fluxes (as proxied by  $\delta^{13}\text{C}_{\text{carb}}$ ) and global climate conditions.  $\delta^{238}\text{U}$  varied in a coordinated, mostly positive relationship with  $\delta^{13}\text{C}_{\text{carb}}$ , indicating that expanded (reduced) oceanic anoxia was linked to lower (higher) productivity. This pattern is inconsistent with productivity control of redox conditions and suggests instead that both proxies responded to a common climatic forcing. We infer that climatic cooling (and glaciation during the Upper Kellwasser Horizon [UKH] event) led to better-ventilated oceanic conditions (higher  $\delta^{238}\text{U}$ ) and greater productivity (higher  $\delta^{13}\text{C}_{\text{carb}}$ ) owing to invigoration of global-ocean overturning circulation and enhanced upwelling. Because the UKH event coincided with the FFB mass extinction, cooling rather than oceanic anoxia may have been the main killing mechanism.

## INTRODUCTION

The biotic crisis at the ca. 372 Ma Frasnian-Famennian boundary (FFB) in the mid–Late Devonian wiped out ~80% of extant marine invertebrate species, making it one of the “Big Five” Phanerozoic mass extinctions (Hallam and Wignall, 1997). Proximate causes of this event are thought to have been expanded marine anoxia (Joachimski and Buggisch, 1993; Formolo et al., 2014; Sim et al., 2015) and/or climatic cooling (Copper, 1986; Joachimski and Buggisch, 2002). However, most redox analyses to date have made use of local proxies, which may differ from mean global-ocean patterns and, thus, not provide an accurate indication of coeval large-scale changes in marine environmental conditions.

Uranium isotopes in marine carbonates represent a new proxy for average global-ocean paleoredox conditions. Its utility is based on (1) a residence time of U in seawater (modern, ~400 k.y.) much longer than the oceanic mixing time (~1–2 k.y.) (Ku et al., 1977), and (2) fractionation of the heavy isotope of U ( $^{238}\text{U}$ ) into reducing facies, resulting in a relative enrichment of seawater in the light isotope ( $^{235}\text{U}$ ) (Weyer et al., 2008; Tissot and Dauphas, 2015). Marine carbonates record contemporaneous seawater  $^{238}\text{U}/^{235}\text{U}$  ratios with little fractionation, although precipitation of secondary carbonates in the sulfate reduction zone can lead to isotopic shifts of +0.2‰ to +0.4‰ (Romaniello et al., 2013). Secular variation

in carbonate  $^{238}\text{U}/^{235}\text{U}$  ratios can be interpreted in terms of changing proportions of anoxic versus oxic sinks for seawater U (Montoya-Pino et al., 2010; Andersen et al., 2014; Dahl et al., 2014; Elrick et al., 2017). Here, we analyze U-isotope variation in a marine carbonate succession through the Frasnian-Famennian (F-F) transition in South China in order to document patterns of mean global-ocean redox variation and their relationship to coeval marine productivity and climate changes.

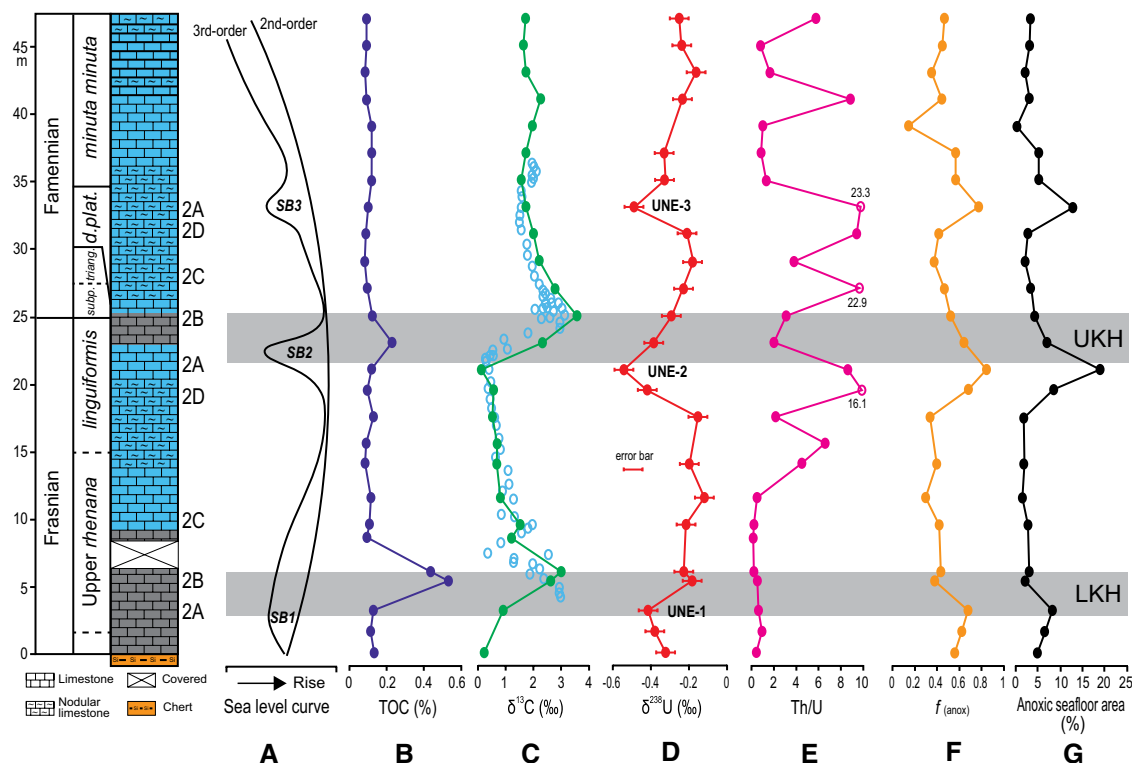
## GEOLOGICAL BACKGROUND

The 47-m-thick study section at Baisha (South China) is located between the cities of Guilin and Wuzhou in Guangxi Province (Fig. DR1A in the GSA Data Repository<sup>1</sup>). During the Late Devonian, it accumulated on the outer margin of the Guilin Platform in the Nanpanjiang Basin, on the eastern margin of the peri-equatorial South China craton, isolated from continental siliciclastic influx (Chen et al., 2005; Fig. DR1B). The FFB interval consists mainly of nodular limestone with intercalated gravity-flow deposits (calciturbidites and gravelly mudflows) and contains the Upper *Palmitolepis rhenana* through *P. minuta minuta* (= Upper *P. triangularis*) conodont zones (Chen and Tucker, 2003). The duration of this interval is 3.5 m.y. (377.7–374.2 Ma; Kaufmann, 2006) to 4.5 m.y. (374.7–370.2 Ma; Becker et al., 2012), yielding an average sedimentation rate of ~10–14 m.m.y.<sup>-1</sup>. The Guilin Platform experienced a broad second-order transgression from the Lower *rhenana* Zone to the top of the *linguiformis* Zone, punctuated by third-order regressions at the base of the Upper *rhenana* Zone, in the uppermost *linguiformis* Zone, and in the Middle *triangularis* Zone (Fig. 1A; Chen and Tucker, 2003). See Item DR1 in the Data Repository for further paleogeographic and conodont zonation information.

The FFB biocrisis is expressed as two discrete events: (1) the Lower Kellwasser Horizon (LKH) in the lower Upper *rhenana* Zone, and (2) the Upper Kellwasser Horizon (UKH) in the uppermost *linguiformis* Zone (Becker et al., 2012). At Baisha, the positions of the UKH at 22–25 m and the FFB at 25 m are unambiguously defined on the basis of conodont data and an ~3‰ positive  $\delta^{13}\text{C}_{\text{carb}}$  excursion (Fig. 1C; Joachimski and Buggisch, 2002; Chen and Tucker, 2003). The position of the LKH is less well constrained owing to an absence of diagnostic conodonts in the lower part of the study section. However, a second ~3‰ positive  $\delta^{13}\text{C}$  excursion at 3–6 m in the Baisha section is likely to represent the LKH event, because no other such feature is known globally from the *rhenana*–*linguiformis*

<sup>1</sup>GSA Data Repository item 2017292, Item DR1 (regional paleogeography and stratigraphy), Item DR2 (analytical methods), Item DR3 (diagenetic influences on study samples), and Item DR4 (marine uranium cycle modeling), is available online at <http://www.geosociety.org/datarepository/2017/> or on request from [editing@geosociety.org](mailto:editing@geosociety.org).

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**Figure 1.** Baisha (South China) section. **A:** Sea-level curve (modified from Chen and Tucker, 2003). **B:** Total organic carbon (TOC). **C:**  $\delta^{13}\text{C}$  (blue points from Chen et al., 2005). **D:**  $\delta^{238}\text{U}$ . **E:** Th/U. Open symbols with numbers show values exceeding the scale maximum. **F:** Modeled U anoxic sink fraction ( $f_{\text{anox}}$ ). **G:** Modeled anoxic seafloor area. SB—sequence boundary; UNE—U-isotope negative excursion; LKH and UKH—Lower and Upper Kellwasser Horizons. Conodont zonation from Ma et al. (2016) and Spalletta et al. (2017): *subp.*—*subperlobata*; *triang.*—*triangularis*; *d. plat.*—*delicatula platys*. Labels 2A–2D refer to panels of Figure 2, which show interpreted marine environmental conditions at the corresponding stratigraphic levels.

interval below the UKH. At Baisha, both of these C-isotope events coincided with sea-level falls and, thus, sequence boundaries (Fig. 1A; Chen and Tucker, 2003).

Similarly to many FFB sections, the LKH and UKH at Baisha show increases in total organic carbon (TOC) relative to background values (Fig. 1B), although the peaks (0.55% and 0.25%, respectively) are smaller than those in Germany (~2%–5%; Joachimski and Buggisch, 1993). Carbonate C-isotopes provide a means of high-resolution correlation of the Baisha section with other FFB sections globally. The two positive excursions in the  $\delta^{13}\text{C}_{\text{carb}}$  profile at Baisha are located in the lower Upper *thenana* Zone and in the uppermost *linguiformis* to lower *subperlobata-triangularis* Zones and correspond to the LKH and UKH, respectively (Fig. 1C). The amplitude of both excursions at Baisha is ~3‰, which is nearly identical to the LKH and UKH excursions in Germany (Joachimski and Buggisch, 1993), Canada (Wang et al., 1996), and Australia (George et al., 2014).

## METHODS

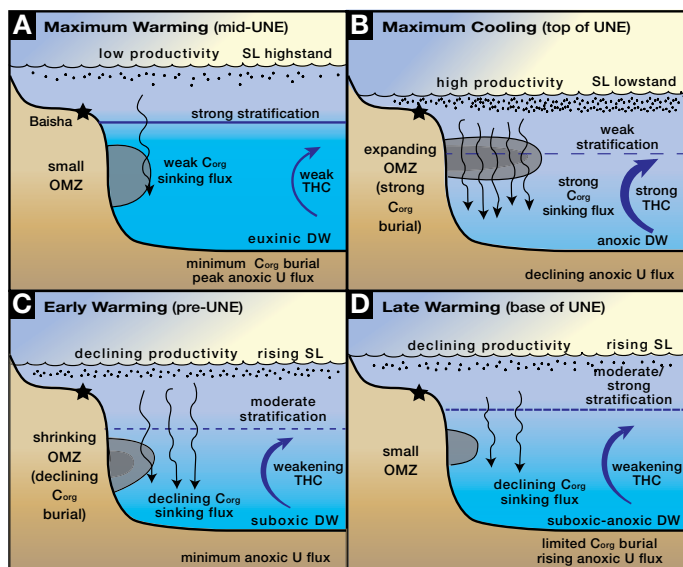
For geochemical analyses, ~3–4 g of powdered carbonate sample was dissolved in 1 M  $\text{HNO}_3$ . Trace-element concentrations in the solution were analyzed using a Thermo inductively coupled plasma quadrupole mass spectrometer (ICP-Q-MS). Uranium isotopes were measured on an aliquot of the solution containing at least 500 ng U. The dissolved material was passed through a UTEVA Resin ([http://www.eichrom.com/products/info/uteva\\_resin.aspx](http://www.eichrom.com/products/info/uteva_resin.aspx)) column, following the procedure outlined by Weyer et al. (2008), to separate U from the matrix. Uranium-isotope measurements were performed on a ThermoFinnigan Neptune multicollector-inductively coupled plasma-mass spectrometer (MC-ICP-MS) using the double-spike MC-ICP-MS procedure described by Weyer et al. (2008). Uranium-isotopic compositions are reported as  $\delta^{238}\text{U}$  ( $^{238}\text{U}/^{235}\text{U}$ ) relative to the U-isotope standard CRM-145. Internal precision (2  $\sigma$ ) of measurements of standards was better than 0.08‰. Average measurement reproducibility (2 $\sigma$ ), assessed by replicate measurements of the same sample solution, is 0.11‰. We monitored data quality through analysis of reference materials BCR-2 (basalt, Columbia River), CRM129a (uranium oxide), and SRM950a (uranium oxide). Carbon isotopes were measured

on a Thermal MAT-253 mass spectrometer, and TOC content on an Eltra 2000 C-S (carbon and sulfur) analyzer. See Item DR2 for further details of analytical methods, and Tables DR1 and DR2 in the Data Repository for full results.

## URANIUM-ISOTOPE RESULTS AND MODELING

The Baisha  $\delta^{238}\text{U}$  profile shows variation from -0.54‰ to 0.00‰ with three negative excursions (Fig. 1D): (1) to -0.41‰ in the lower Upper *thenana* Zone (0–2 m; U-isotope negative excursion 1 [UNE-1]); (2) to -0.54‰ in the uppermost *linguiformis* to lower *subperlobata-triangularis* Zones (19–25 m; UNE-2); and (3) to -0.49‰ in the upper *delicatula platys* Zone (33–35 m; UNE-3). Th/U ratios range from <1 to 24 and tend to increase in conjunction with lower  $\delta^{238}\text{U}$  values (Fig. 1E), although the former show greater sample-to-sample variation. Comparisons of U-system data to  $\delta^{18}\text{O}$ , Mg/Ca, Sr/Ca, and [Al] revealed no systematic diagenetic effects (see Item DR3 for details). The three third-order regressions identified through lithofacies analysis (Chen and Tucker, 2003) correspond closely to the three negative uranium isotope excursions (UNE-1 to UNE-3) identified in the present study. Based on average sedimentation rates, the cycles culminating in UNE-2 and UNE-3 had durations of ~1.3–2.0 m.y. and 0.7–1.0 m.y., respectively.

We employed a mass balance reservoir model (from Montoya-Pino et al. [2010], as modified by Elrick et al. [2017]) to quantitatively estimate the flux of U into anoxic facies and associated changes in global-ocean redox conditions through the F-F transition (see Item DR4 and Table DR2 for model details and output). The results suggest that the anoxic sink flux expanded to ~0.68 during UNE-1, decreased to ~0.3–0.4 before peaking at ~0.84 during UNE-2, and finally fluctuated from 0.1 to 0.5 following UNE-2 with a peak at ~0.78 during UNE-3 (Fig. 1F). Uranium anoxic sink fluxes can be converted to anoxic seafloor area using a transfer function (Tissot and Dauphas, 2015), yielding estimates of 8%, 20%, and 12% for UNE-1, UNE-2, and UNE-3, respectively (versus 0.2% for the modern ocean; Fig. 1G). Note that measured  $\delta^{238}\text{U}$  values were adjusted by 0.4‰ to account for fractionation related to secondary carbonate formation in euxinic pore waters (Table DR2; Romaniello et al., 2013); uncertainties



**Figure 2.** Marine environmental conditions within  $\delta^{13}\text{C}$ - $\delta^{238}\text{U}$  cycles of Frasnian-Famennian (F-F) transition. **A:** Maximum warming (mid-UNE [U-isotope negative excursion]). **B:** Maximum cooling (top of UNE). **C:** Early warming (pre-UNE). **D:** Late warming (base of UNE). **A** is equivalent to onset of Lower Kellwasser Horizon (LKH) and Upper Kellwasser Horizon (UKH), and **B** is equivalent to their terminations. See text for discussion. SL—sea level; OMZ—oxygen minimum zone;  $\text{C}_{\text{org}}$ —organic carbon; DW—deep water; THC—thermohaline circulation].

in this correction factor will affect absolute values of anoxic sink flux and seafloor area but not their secular variation patterns.

The U- and C-isotope profiles of Baisha covary: each negative  $\delta^{238}\text{U}$  excursion marks the end of a protracted decline (of  $\sim 2\text{‰}$ – $3\text{‰}$ ) and the start of a rapid rise (also  $\sim 2\text{‰}$ – $3\text{‰}$ ) in  $\delta^{13}\text{C}_{\text{carb}}$  values, a pattern that is especially well developed between the UNE-1 and UNE-3 horizons (Fig. 1). For UNE-1, only the sharp rise is seen because the study section extends only to the base of this horizon. Above UNE-3, C-isotopic variation becomes muted, with only a small rise ( $\sim 0.5\text{‰}$ ) followed by nearly uniform values.

## MARINE PRODUCTIVITY-REDOX RELATIONSHIPS DURING THE F-F TRANSITION

The sharp positive  $\delta^{13}\text{C}_{\text{carb}}$  excursions marking the LKH and UKH in Germany coincided with  $\delta^{18}\text{O}_{\text{phosphate}}$  excursions of  $\sim +1.5\text{‰}$  to  $+2.0\text{‰}$  (Joachimski and Buggisch, 2002). Shifts of this magnitude reflect substantial tropical sea-surface cooling ( $\sim 6$ – $8^\circ\text{C}$ ), continental ice mass expansion (equivalent to  $\sim 150$ – $200$  m of eustatic fall), or some combination of these influences. Significantly, FFB shifts in  $\delta^{13}\text{C}_{\text{carb}}$  profiles slightly lead those in  $\delta^{18}\text{O}_{\text{phosphate}}$  profiles (Joachimski and Buggisch, 2002, their figure 1), suggesting that C-cycle perturbations drove temperature changes during the FFB event. The link between higher  $\delta^{13}\text{C}_{\text{carb}}$  and cooling events was probably through invigoration of oceanic overturning circulation, enhancing upwelling of nutrients and surface-water productivity. Cooling during the UKH event was sufficient to trigger a brief episode of continental glaciation (Streel et al., 2000) and a eustatic fall at the FFB (Johnson et al., 1985; Fig. 1A) that led to widespread karst development on shallow carbonate platforms (Chen and Tucker, 2004; Chow et al., 2004). The evidence for cooling and continental glaciation during the UKH event is sufficiently strong that we will interpret  $\delta^{13}\text{C}_{\text{carb}}$ - $\delta^{238}\text{U}$  variation at Baisha in this context.

The large ( $2\text{‰}$ – $3\text{‰}$ ) oscillations in  $\delta^{13}\text{C}_{\text{carb}}$  through the F-F transition were probably related to changes in organic carbon burial fluxes driven by changes in marine productivity (George et al., 2014). In this scenario, positive  $\delta^{13}\text{C}_{\text{carb}}$  shifts reflect preferential removal of  $^{12}\text{C}$  to the sediment

via elevated levels of marine productivity. Other possible causes for rapid positive shifts are few, in contrast to rapid negative  $\delta^{13}\text{C}_{\text{carb}}$  shifts, which have multiple possible sources (e.g., volcanic  $\text{CO}_2$  emissions, oxidation of seafloor clathrates or permafrost methane, and soil erosion events; Berner, 2002).

The observation that the same pattern of  $\delta^{13}\text{C}_{\text{carb}}$ - $\delta^{238}\text{U}$  covariation is repeated three times through the study section implies coordinated responses of marine productivity and redox conditions to a cyclic forcing mechanism (cf. George et al., 2014). The bases of both the LKH and UKH exhibit simultaneous minima of  $\delta^{238}\text{U}$  and  $\delta^{13}\text{C}_{\text{carb}}$ , indicating that the most reducing conditions at Baisha were associated with limited productivity. This pattern is not consistent with a productivity control on oceanic redox (when higher productivity would lead to more reducing conditions) but, rather, consistent with responses of both proxies to a common climate forcing, i.e., warming, which led to water-column stratification, reduced productivity, and intensified deep-water anoxia (Fig. 2A). However, the durations of these cycles (from 0.7 to 2.0 m.y.) probably preclude control of climatic variation through Milankovitch orbital forcing.

Within the  $\sim 3$  m intervals of both the LKH and UKH,  $\delta^{238}\text{U}$  and  $\delta^{13}\text{C}_{\text{carb}}$  rise rapidly, suggesting concurrent shifts toward less-reducing and higher-productivity conditions. Again, this is inconsistent with a productivity control on redox conditions but consistent with a common climatic forcing (e.g., cooling, leading to invigorated oceanic overturning circulation, increased upwelling and productivity, and improved deep-water ventilation; Fig. 2B). Because these intervals coincided with mass extinctions (Ma et al., 2016), it implies that the biocrises were due not to oceanic anoxia (which was already attenuating following a peak at the bases of the LKH and UKH) but, rather, to rapidly cooling oceanic temperatures (cf. Copper, 1986; Joachimski and Buggisch, 2002).

Although  $\delta^{13}\text{C}_{\text{carb}}$  reaches peaks at the top of both the LKH and UKH,  $\delta^{238}\text{U}$  levels off or continues to rise weakly (Fig. 1D), suggesting a modest decoupling of redox and productivity conditions during this stage of each cycle. One possibility is that this was an interval of relatively stable climate and ocean ventilation (as suggested by limited  $\delta^{238}\text{U}$  variation), and the trend toward reduced productivity (lower  $\delta^{13}\text{C}_{\text{carb}}$ ) reflects a decline in deep-water nutrient inventories over time. However, the steep negative shift of  $\delta^{238}\text{U}$  toward the top of these intervals (indicating a rapid onset of more widely reducing conditions) belies any interpretation of climatic-oceanic stability. Instead, we interpret the observed patterns as slowly warming intervals, leading gradually to intensified water-column stratification and reduced upwelling and productivity (Fig. 2C). The initial lack of response in the  $\delta^{238}\text{U}$  profile suggests that these environmental changes did not have an immediate effect on global-ocean redox conditions, perhaps because simultaneous intensification of water-column stratification and reduction of organic-matter sinking fluxes exerted opposite, and largely balancing, influences on redox conditions.

The comparatively abrupt onset of negative  $\delta^{238}\text{U}$  excursions suggests a rapid expansion of oceanic anoxia. We infer that once the gradual warming reached a critical threshold, stratification was sufficiently intense to trigger widespread deep-water anoxia, increasing the anoxic sink flux of U (i.e., a rapid shift toward lower  $\delta^{238}\text{U}$ ) (Fig. 2D). It should be noted that U and  $\text{C}_{\text{org}}$  (organic carbon) burial fluxes are not tightly coupled at this stage. Initially, expansion of deep-water anoxia had little effect on organic carbon burial fluxes because marine productivity was generally low. However, as anoxia persisted, larger amounts of organic matter began to accumulate in anoxic facies, drawing down atmospheric  $\text{CO}_2$  levels and initiating the next cooling trend.

Evidence for expanded oceanic anoxia during the UKH event based on local redox proxies such as trace-metal concentration data and nitrogen isotopes (Racki et al., 2002; Whalen et al., 2015) does not conflict with our results. Indeed, it is consistent with predictions of general ocean-circulation models for intervals of invigorated thermohaline circulation as a result of global climatic cooling (e.g., Haywood et al., 2000). This

process would lead to productivity increases, expansion of anoxia, and generally unsettled marine environmental conditions in upwelling zones. The importance of our study is that carbonate  $\delta^{238}\text{U}$  is a global-ocean redox proxy that provides a spatially integrated signal transcending local effects in regions of upwelling.

## CONCLUSIONS

The FFB section at Baisha, South China, exhibits coordinated cyclicity of  $\delta^{13}\text{C}_{\text{carb}}$  and  $\delta^{238}\text{U}$ , which proxy secular variation in mean global-ocean marine productivity and redox conditions, in response to a common climate forcing. Climate warming led to water-column stratification, reduced productivity (low  $\delta^{13}\text{C}_{\text{carb}}$ ), and intensified deep-water anoxia (low  $\delta^{238}\text{U}$ ). Strong climatic cooling during the LKH and UKH events led to invigorated oceanic overturning circulation, increased upwelling and productivity (high  $\delta^{13}\text{C}_{\text{carb}}$ ), and improved deep-water ventilation (high  $\delta^{238}\text{U}$ ). Because the UKH event coincided with a mass extinction, cooling rather than oceanic anoxia is likely to have been the main killing mechanism.

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